

Introduction

Modern translators enable easy communication among different language users. Even though these translators can translate sign languages to texts, barriers exist for Deaf, Mute, and Hard-of-Hearing population using different sign languages. The current solution for different sign language users to communicate requires three-step translations. In our project, we develop and evaluate the accuracy of a direct interactive **sign2sign** language translation using deep learning models. In this poster, we present our preliminary results of translations of ten gestures between American and Chinese Sign Languages.

Research Question

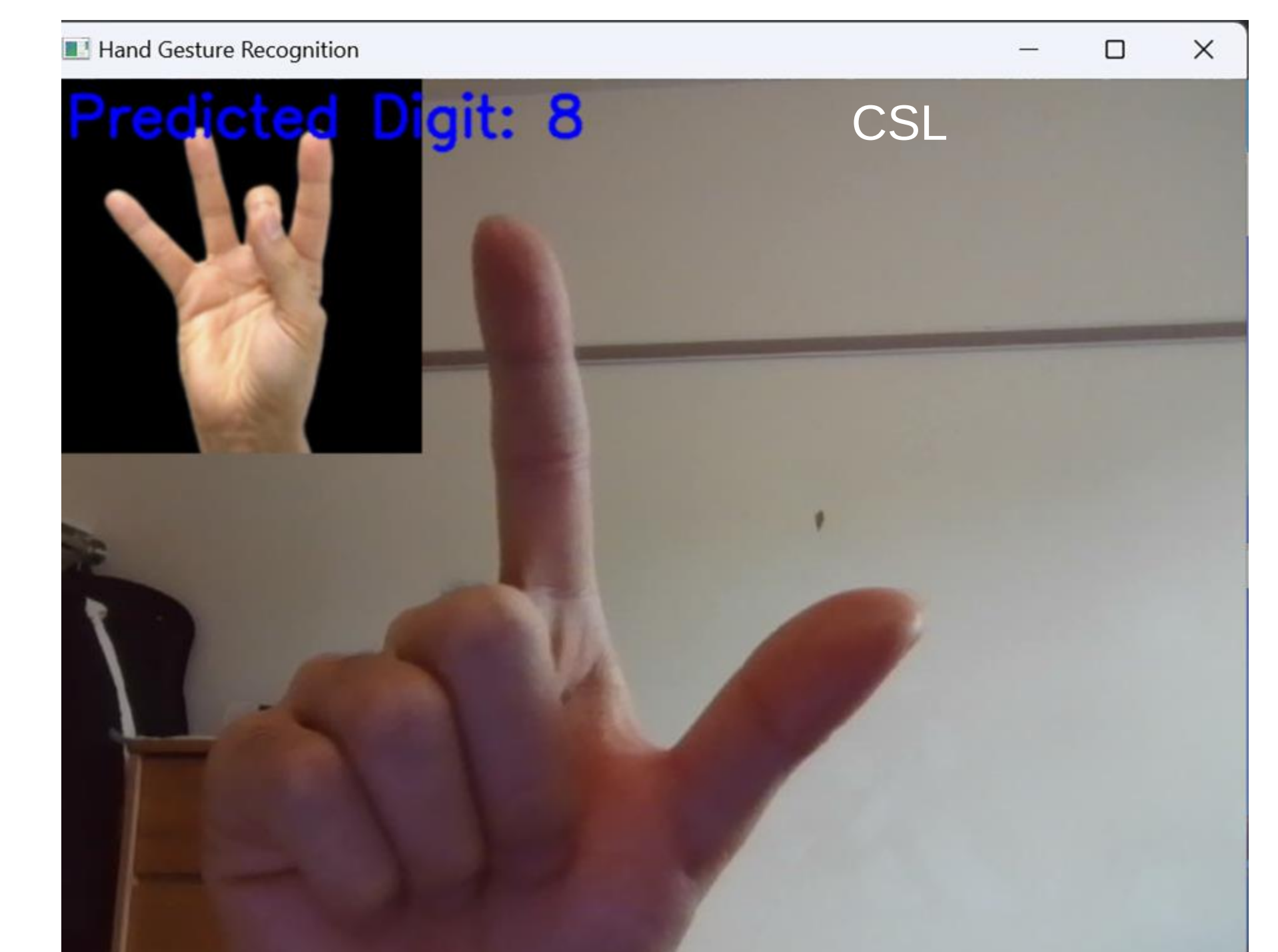
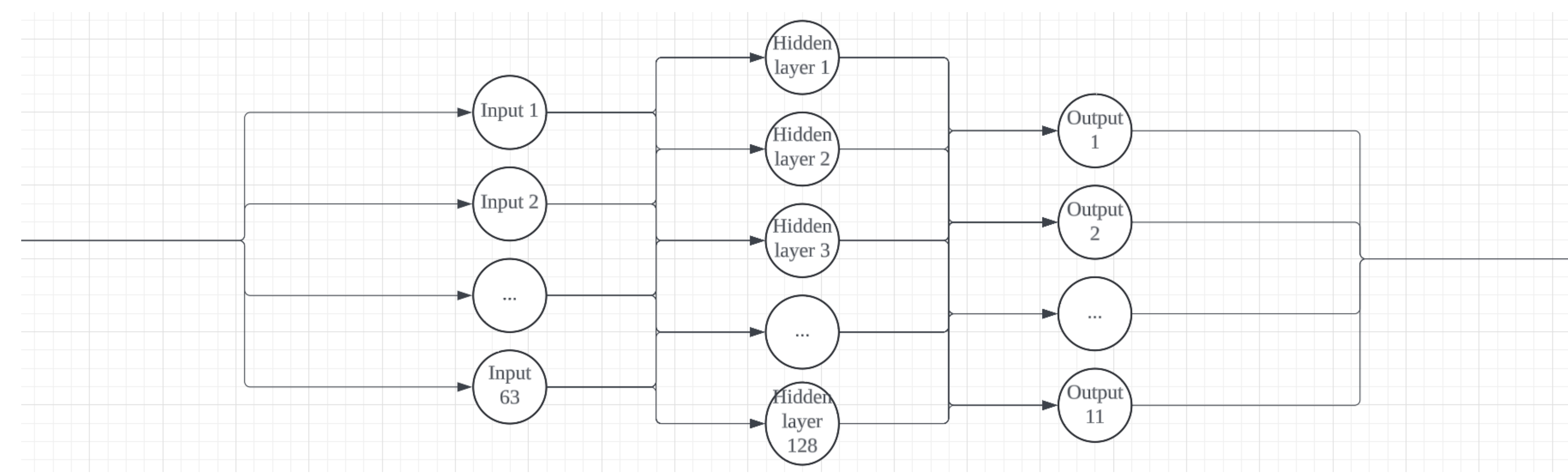
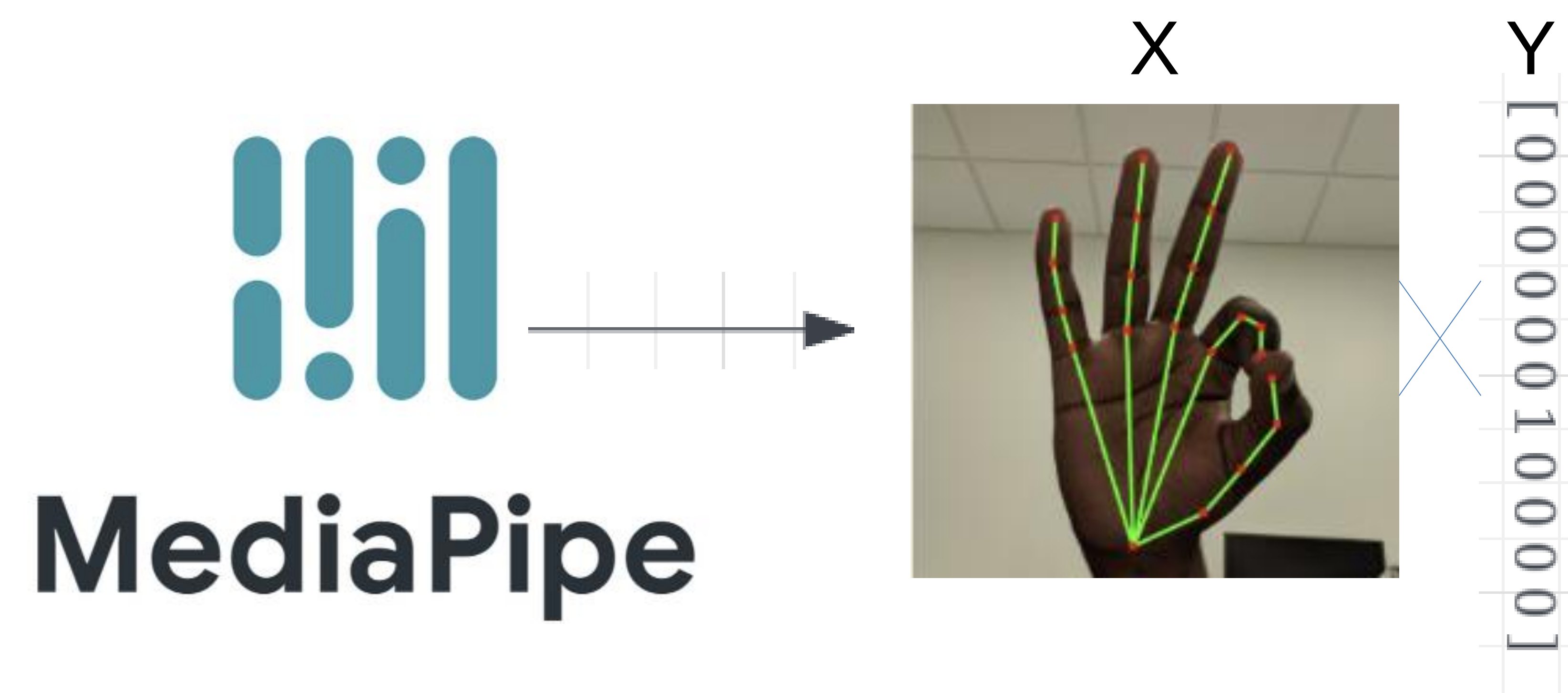
How accurate to directly translate between two sign languages in real-time using contemporary computer vision and deep learning techniques?

Research Value

- Develop a low cost sign language to sign language translation system
- Evaluate the accuracy of a direct translation approach
- Allow Deaf and Muted communities over the globe communicate without using text as a medium



Image illustration generated using ChatGPT4.



Methods

- 14K total images (7K each)
- 80% used for training and 20% for testing
- Ground-truth: a probability vector of size 11. One of each gesture and an additional one for non-digit. e.g. [1, 0, 0, ..., 0] for gesture 1.

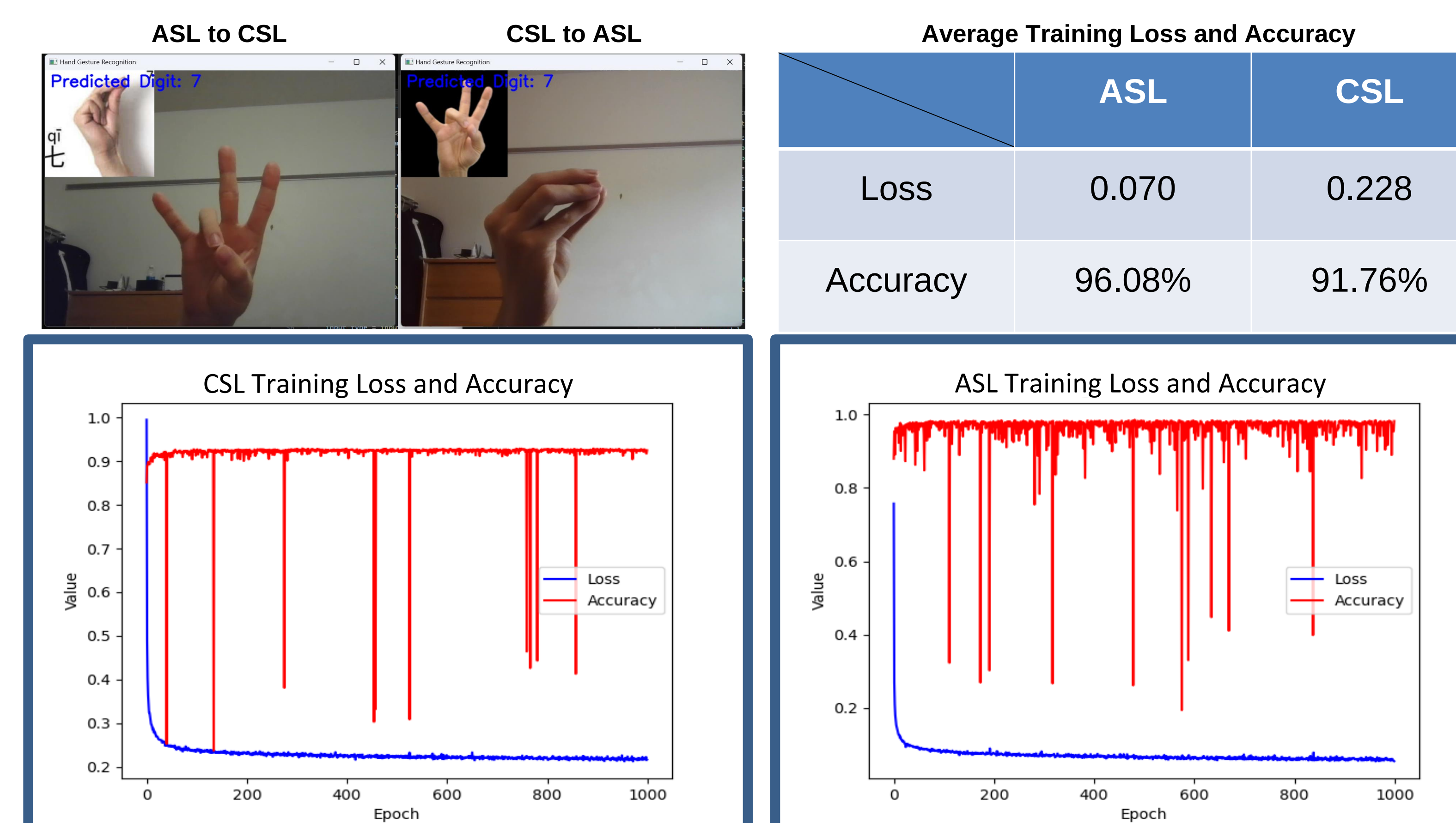
- Prediction: Given a prediction vector:

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Predicted Digit: 8
Probabilities: [[1.00425456e-23 0.00000000e+00 0.00000000e+00 0.00000000e+00
6.14791675e-41 9.52882956e-44 0.00000000e+00 1.00000000e+00
8.21408337e-15 1.15128246e-23 0.00000000e+00]]
```

The predicted gesture has the **highest probability**.

- Accuracy: $\frac{\text{Number of correct predictions}}{\text{Total predictions}}$

Preliminary Results



Future Work

- Extend to words and sentences
- Evaluate different deep learning models (e.g. LLM)
- Implement an Augmented Reality interface

Reference

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